LEVEL SETS OF A WEIGHTING ARE LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22520-22575, immediately following the statement at 22513-22518.

Editorial note. The proof contains a footnote about immersions and nonzero loci; it is retained verbatim inside the span.

Editorial note. Five cross-chapter pointers (Algebra, Morphisms, Properties, Topology) are recorded in assembly.external_references.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 22528 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Openness of the locus where a weighting is nonzero. chapter lemma lemma-weighting-open, used from proof line 22521 without further mention. Gives openness of the nonzero locus, which is the basic input for constructibility of level sets.
- Base change of weightings. chapter lemma lemma-weighting-base-change, used from proof line 22521. Lets the argument be carried out after changing the base.
- Local constructibility and constructible subsets. Topology, section topology-section-constructible and Properties, section properties-section-constructible, cited at proof lines 22524-22526. Supplies the closure properties of locally constructible sets used to combine the local pieces.
- Chevalley's theorem and massaging finite morphisms. Morphisms, theorem morphisms-theorem-chevalley and lemma morphisms-lemma-massage-finite, cited in the proof. Makes images of constructible sets constructible and normalises a finite morphism into a disjoint union of finite locally free pieces.
- Passing through finite morphisms. chapter lemma lemma-quasi-finite-separated-pass-through-finite, cited in the proof. Reduces the locally quasi-finite situation to a finite one.

Source reference. Original reference 00F8: focused-seeds/geometry/S9-morph-jumps-weighting-level-sets/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05K0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12457–12457. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22173–22173. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GK8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22239–22239. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HW: focused-seeds/geometry/S9-morph-jumps-weighting-level-sets/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 054K: focused-seeds/geometry/S9-morph-jumps-weighting-level-sets/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 054B: focused-seeds/geometry/S9-morph-jumps-weighting-level-sets/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04ZC: focused-seeds/geometry/S9-morph-jumps-weighting-level-sets/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism. Let $w : X \to \mathbf{Z}$ be a weighting of f. Then the level sets of the function w are locally constructible in X.*

Proof. In the proof below we will use Lemmas 0GK8 and 0F3B without further mention. We will also use elementary properties of constructible subsets of schemes and topological spaces, see Topology, Section 04ZC and Properties, Section 054B. Using this the reader sees question is local on X and Y; details omitted. Hence we may assume X and Y are affine. If we can find a surjective morphism $Y' \to Y$ of finite presentation such that the level sets of w pull back to locally constructible subsets of $X' = Y' \times_Y X$, then we conclude by Morphisms, Theorem 054K.

Assume X and Y affine. We may choose an immersion $X \to T$ where $T \to Y$ is finite, see Lemma 05K0. By Morphisms, Lemma 03HW after replacing Y by Y' surjective finite locally free over Y, replacing X by $Y' \times_Y X$ and T by a scheme finite locally free over Y' containing $Y' \times_Y T$ as a closed subscheme, we may assume T is finite locally free over Y, contains closed subschemes T_i mapping isomorphically to Y such that $T = \bigcup_{i=1,\ldots,n} T_i$ (set theoretically). Since $T_i \subset T$ is a constructible closed subset (as the image of a finitely presented morphism $Y \to T$ of schemes), we

see that for $I \subset \{1, \ldots, n\}$ the intersection $\bigcap_{i \in I} T_i$ is a constructible closed subset of T and hence maps to a constructible closed subset of Y.

For a disjoint union decomposition $\{1, \ldots, n\} = I_1 \amalg \ldots \amalg I_r$ with nonempty parts consider the subset $Y_{I_1,\ldots,I_r} \subset Y$ consisting of points $y \in Y$ such that $T_y = \{x_1, \ldots, x_r\}$ consists of exactly r points with $x_j \in T_i \Leftrightarrow i \in I_j$. By our remarks above this is a constructible partition of Y. There exists an affine scheme Y' of finite presentation over Y such that the image of $Y' \to Y$ is exactly $Y_{I_1,\ldots,I_r}$, see Algebra, Lemma 00F8. Hence we may assume that $Y = Y_{I_1,\ldots,I_r}$ for some disjoint union decomposition $\{1, \ldots, n\} = I_1 \amalg \ldots \amalg I_r$. In this case $T = T(1) \amalg \ldots \amalg T(r)$ with $T(j) = \bigcap_{i \in I_j} T_i$ is a decomposition of T into disjoint closed (and hence open) subsets. Intersecting with the locally closed subscheme X we obtain an analogous decomposition $X = X(1) \amalg \ldots \amalg X(r)$ into open and closed parts. The morphism $X(j) \to Y$ an immersion. Since w is a weighting, it follows that $w|_{X(j)}$ is locally constant[1] and we conclude. $\square$

[1]In fact, if $f : X \to Y$ is an immersion and w is a weighting of f, then f restricts to an open map on the locus where w is nonzero.